

Syllabus

Spring semester 2026

Course Title	Electrochemical Energy System	Course Code		ECB4107	Credit	3	Classroom	Hyung3207
		Type		Chemistry	Hours	3		
Teacher	Name	Class no.	Office	Office Hour*	Contact		E-mail	
	Gonu Kim	1	Chang-603	Wed. 3, Fri. 3, Fri. 4	051-606-2330		gonukim82@ksa.kaist.ac.kr	

*Office Hour: Time for students to ask questions and discuss lecture or lab topics with teachers for deeper understanding.

1. Course Descriptions

Course Objectives
1) This course introduces the concepts used in electrochemistry to students who have learned basic chemistry theory. 2) Students will learn the principles and applications of electrochemical techniques used in real life. 3) Students will increase their creative thinking, logic through presentations and discussion classes
Contents
This course focuses on the chemical theory and application of electrical energy, the most convenient form of energy that is closely related to everyday life. Explanations will be made on basic physical chemistry of electrochemistry and the representative applications such as batteries, solar cells, fuel cells, electrolysis. -thermodynamics and kinetics in electrochemistry, electrode potentials, galvanic and electrolytic cells, mass transport by migration, diffusion and convection. Conductivity. The electrochemical double layer. Overpotentials. Electrochemical instrumentation and techniques. Cyclic voltammetry, Electrochemical impedance spectroscopy. Intercalation processes. Corrosion. Mixed potentials. Polarisation curves.

2. Core Learning Competencies

Subject-Specific Core Competencies						
Content Comprehension	Scientific Thinking	Critical Thinking	Problem-Solving Skills	Communication Skills	Collaboration Skills	Creativity
This course aims to cultivate students' content comprehension, scientific thinking, and critical thinking through lectures, group presentations, and individual presentations. In addition, collaborative leadership and creativity are fostered through group-based learning activities.						

3. Textbooks and References

A. Textbook: Instructor-developed handouts

B. References

1: Seung Mo Oh, Electrochemisty, 3rd Ed., 자유아카데미

2: Bard and Faulkner, Electrochemical method, 2nd Ed., Wiley 2001

4. Teaching Methods

This course is conducted using instructor-developed materials and follows a flipped-learning and discussion-centered format, emphasizing review, discussion, and presentations.

A. Students are required to study the provided handouts prior to class to acquire foundational concepts.

B. Class sessions focus on group learning, discussions, Q&A sessions, and presentations.

5. Assessment Criteria and Methods

A. Assessment Breakdown

Group Presentation	Individual Presentation	Assignments	Final Examination	Attendance & Participation	Total
20%	20%	10%	40%	10%	100%

B. Assessment Methods

(1) Group and Individual Presentations

- Group Presentation: Students select and present topics related to electrochemistry (e.g., lithium-ion batteries, solar cells, fuel cells, LEDs).

- Individual Presentation: Students present on recent research papers in the field of electrochemistry.

(2) Assignments

- Assignments related to course content are discussed in groups of 3–4 students.

- Class preparedness and participation in group discussions are evaluated.

- If participation is insufficient, 0.5 to 1 point may be deducted per session.

(3) Written Examination-

- One final examination will be administered.

- Exam questions are based on course content and consist primarily of short-answer and descriptive questions.

(4) Attendance and Participation

- Disruptive behaviors such as chatting, sleeping, or engaging in other coursework during class will result in a 0.5-point deduction from the participation score.
- Active and positive contributions to group learning may earn a 0.5-point bonus.
- Late arrival: 0.5-point deduction per occurrence.
- Unexcused absence: 1-point deduction per absence.
- Four or more unexcused absences will result in a Fail grade.

6. Credit Awarding Policy

☞ Students demonstrating excellent academic achievement may be recommended for a Course Excellence Award. Their course records will include core competencies, detailed descriptions of academic performance, assessment results, participation, and learning attitude. Students with poor attendance or unsatisfactory learning attitudes will be excluded from recommendation.

☞ For students who miss presentation assessments:

- Excused absence: 100% credit
- Medical absence: 80% credit

※ **Adjusted Score = Maximum Presentation Score × (Written Exam Score / Maximum Written Exam Score) × Adjustment Rate**

※ **The adjustment rate follows the Academic Performance Management Regulations.**

7. Weekly Plans

Week (period)	Topic and Contents	(Notes)
Week 1 2.23.~3.1.	Fundamentals of electrochemistry -Electrical quantities	
Core Competencies	Content Comprehension, Scientific Thinking Skills, Critical Thinking Skills, Problem-Solving Skills, Communication Skills, Collaboration Skills, and Creativity	
Week 2 3.2.~3.8.	Fundamentals of electrochemistry -Ions in solution: electrolyte	3.2.
Core Competencies	Content Comprehension, Scientific Thinking Skills, Critical Thinking Skills, Problem-Solving Skills, Communication Skills, Collaboration Skills, and Creativity	Substitute Holiday
Week 3 3.9.~3.15.	Charge separation at interface - Electric double layer, Space charge region, Diode and transistor	
Core Competencies	Content Comprehension, Scientific Thinking Skills, Critical Thinking Skills, Problem-Solving Skills, Communication Skills, Collaboration Skills, and Creativity	
Week 4 3.16.~3.22.	Thermodynamics and kinetics in electrochemistry - Electrode reaction rate	

Core Competencies	Content Comprehension, Scientific Thinking Skills, Critical Thinking Skills, Problem-Solving Skills, Communication Skills, Collaboration Skills, and Creativity	
Week 5 3.23.~3.29.	Electrochemical cells - Galvanic cells, Electrolytic cells	
Core Competencies	Content Comprehension, Scientific Thinking Skills, Critical Thinking Skills, Problem-Solving Skills, Communication Skills, Collaboration Skills, and Creativity	
Week 6 3.30.~4.5.	Group presentations (lithium-ion battery) - Operation principle, efficiency, advantages and disadvantages	
Core Competencies	Content Comprehension, Scientific Thinking Skills, Critical Thinking Skills, Problem-Solving Skills, Communication Skills, Collaboration Skills, and Creativity	
Week 7 4.6.~4.12.	Group presentations (solar cell) - Operation principle, efficiency, advantages and disadvantages	
Core Competencies	Content Comprehension, Scientific Thinking Skills, Critical Thinking Skills, Problem-Solving Skills, Communication Skills, Collaboration Skills, and Creativity	
Week 8 4.13.~4.19.	Midterm Exams (4.14.~4.17.)	4.13. Regular Classes

Week 9 4.20.~4.26.	Group presentations (fuel cell) - Operation principle, efficiency, advantages and disadvantages	
Core Competencies	Content Comprehension, Scientific Thinking Skills, Critical Thinking Skills, Problem-Solving Skills, Communication Skills, Collaboration Skills, and Creativity	
Week 10 4.27.~5.3.	Group presentations (light emitting diode) - Operation principle, efficiency, advantages and disadvantages	
Core Competencies	Content Comprehension, Scientific Thinking Skills, Critical Thinking Skills, Problem-Solving Skills, Communication Skills, Collaboration Skills, and Creativity	
Week 11 5.4.~5.10.	Cyclic voltammetry	5.5. Children's Day
Core Competencies	Content Comprehension, Scientific Thinking Skills, Critical Thinking Skills, Problem-Solving Skills, Communication Skills, Collaboration Skills, and Creativity	
Week 12 5.11.~5.17.	Electrochemical impedance spectroscopy	
Core Competencies	Content Comprehension, Scientific Thinking Skills, Critical Thinking Skills, Problem-Solving Skills, Communication Skills, Collaboration Skills, and Creativity	

Week 13 5.18.~5.24.	Individual presentations - (papers related to electrochemistry subjects)	
Core Competencies	Content Comprehension, Scientific Thinking Skills, Critical Thinking Skills, Problem-Solving Skills, Communication Skills, Collaboration Skills, and Creativity	
Week 14 5.25.~5.31.	Individual presentations - (papers related to electrochemistry subjects)	5.25. Substitute Holiday
Core Competencies	Content Comprehension, Scientific Thinking Skills, Critical Thinking Skills, Problem-Solving Skills, Communication Skills, Collaboration Skills, and Creativity	
Week 15 6.1.~6.7.	Individual presentations - (papers related to electrochemistry subjects)	6.3. Local Election Day
Core Competencies	Content Comprehension, Scientific Thinking Skills, Critical Thinking Skills, Problem-Solving Skills, Communication Skills, Collaboration Skills, and Creativity	
Week 16 6.8.~6.14.	Make-up class week	
Week 17 6.15.~6.19.	Final Exams (6.18.~6.19.)	

※ The above plans may change according to the academic schedule.

[Reference] Syllabus Competence Reference

1. Analytical and Creative Thinking	2. Complex Communication	3. Leadership and Teamwork	4. Digital/Quantitative Literacy	5. Global Perspective	6. Adaptability/Progression /Spirit of Adventure	7. Integrity and Ethical Decision Making	8 Habits/Mindset
Ability to identify, manage and deal with complex problems	Ability to understand and express themselves in two or more languages	Ability to be the first to come up with and present new ideas	Ability to understand, use and apply digital technology	Ability to develop an open mind about the traditions or values of others	Ability to develop flexibility, agility and adaptability	Ability to sustain an empathetic and understanding perspective	Sincerity
Ability to see through prejudice and identify reliable information and uncertain information	Ability to communicate clearly with diverse audiences or readers	Ability to influence people to follow themselves, rather than lead them through directives, orders, or authority	Ability to create digital contents and media	Ability to understand the history, politics, religion, and culture of another region	Ability to challenge unfamiliar situations	Ability to cultivate sincerity, honesty, fairness, and respect creativity	Creativity
Ability to find quality information in the flood of information	Ability to listen attentively	Ability to build trust, resolve conflicts, and support others	Ability to utilize multimedia resources to effectively communicate one's ideas in a variety of formats	Ability to use one or more of the international languages used worldwide	Ability to explore and experiment	Ability to display moral courage in the face of injustice	A love of learning and curiosity
Ability to articulate meaningful questions	Ability to speak effectively and coherently	Ability to facilitate group discussions or discussions, build consensus, and negotiate outcomes	Ability to master and use higher mathematics	Ability to use technology to connect people and things around the world	Ability to word effectively in situations of ambiguous or changing priorities	Ability to act responsibly for the benefit and happiness of the greater community	Resilience/elasticity
Ability to analyze and create ideas or knowledge	Ability to write concisely and clearly, considering the diversity of readers	Ability to teach, guide, and counsel Ability to seek help from others	Ability to understand traditional or recently emerging fields such as mathematics, science, technology, environmental science, robotics, geometry, cellular automata, nanotechnology, and biotechnology	Social and intellectual skills to effectively navigate information across multiple cultures	Ability to see failure as a lesson and to accept that innovation is made up of frequent failures and small successes	Ability to have an intrinsic understanding of ethical issues or dilemmas about new media or new technologies	Perseverance/patience
Ability to devise and test solutions to problems by	Ability to explain information and persuasively	Ability to collaborate on tasks, manage groups, and		Ability to use 21 st century core competencies such as communication	Ability to develop a sense of independence to seek new roles,	Ability to make rational and ethical decisions on complex	Self-efficacy

1. Analytical and Creative Thinking	2. Complex Communication	3. Leadership and Teamwork	4. Digital/Quantitative Literacy	5. Global Perspective	6. Adaptability/Progression/Spirit of Adventure	7. Integrity and Ethical Decision Making	8 Habits/Mindset
trial and error	engage others in an engaging way	delegate responsibilities		skills, collaboration skills, critical thinking skills, creativity to properly understand and deal with global issues	ideas, or strategies	issues	
Ability to think of different ways to cope		Ability to listen and practice goals		Ability to learn from and collaborate with people from different cultures, religions and lifestyle through mutual respect and open dialogue	Ability to develop the knowledge and competencies necessary to start a business and run a business		Stress management skills
Ability to develop knowledge and perspectives that are not limited to one field but span multiple fields		Ability to share credit for success		Ability to generate diverse ideas and exploit social and cultural differences to get things done			Time management skills
Ability to make sustained inferences							
Ability to put things together or adapt them to new uses and situations							
Ability to solve new problems that cannot be solved through established rules							
Ability to use knowledge or creativity to solve complex real-world problems							

1. Complex Problem Solving Skills	Ability to solve new and unclear problems that may arise in complex real-world situations
2. Critical Thinking Skills	Ability to compare and analyze the strengths and weaknesses of various solutions or approaches to a given problem using logic and reasoning
3. Creativity	Ability to come up with novel ideas when given a certain topic or situation or to suggest a creative method in the process of solving a problem
4. Human Resource Management Skills	Ability to select the most suitable person for a specific task and manage them to achieve maximum work performance through motivation, systematic and efficient management
5. Ability to Collaborate	Ability to adjust one's own behavior in response to the behavior of others
6. Emotional Ability	Ability to perceive other people's reactions and understand why they react the way they do
7. Judgement and Decision-making Skills	Ability to make the most rational judgement and decision by analyzing the costs and benefits of a specific situation or problem
8. Orientated Towards Service	Ability to actively seek ways to help others
9. Negotiation Skills	Ability to bring together parties who differ in ideas or interests and try to reduce those differences
10. Cognitive Flexibility	Ability to apply different kinds of rules or principles to rearrange and reorganize things in different ways